instance since this process is really "real-time re-optimization" and not true mathematical "dynamic optimization." But regardless of the nomenclature used, the goal is for investors to define how the algorithm will react to changing market conditions.

There are often times when investors may not want the algorithm to adapt to changing market conditions. For example, investors seeking to achieve the day's closing price would not want to make any adjustments to the algorithm because it may cause the algorithm to finish early and increase tracking error compared to the closing price. Investors seeking to achieve the VWAP price would want to adhere to the intraday volume profile regardless of price movement or volatility. Additionally, investors trading hedged baskets or hedged portfolios may not want to deviate from their initial prescribed strategy regardless of market conditions since doing so may ruin the hedge and increase risk exposure. Portfolio adaptation tactics are further discussed in Chapter 9, Portfolio Algorithms.

Below we discuss three methodologies for revising intraday algorithmic trajectories based on expected total trading cost. These are: targeted cost, aggressive-in-the-money (AIM), and passive-in-the-money (PIM) strategies[1]. These studies also provide an in-depth analysis surrounding the underlying profit and loss distributions corresponding to these tactics as well as real-time solutions.

Adaptation tactics and how they influence algorithmic decision and trading performance were previously studied by Kissell and Malamut (2005, 2006) and Almgren and Lorenz (2006, 2007). We follow the approach from the Journal of Trading introduced by Kissell and Malamut (2005, 2006).

Projected Cost

At the beginning of trading the projected cost is equal to the initial cost estimate (described in step 2 above). But after trading begins the projected cost will be comprised of four components: realized cost, momentum cost, remaining market impact cost, and alpha trend cost.

[1]Tom Kane, former Managing Director at JP Morgan and Merrill Lynch, introduced the naming of the Aggressive-in-the-Money (AIM) and Passive-in-the-Money (PIM) adaptation tactics. Furthermore, unlike many of the names chosen for algorithms, the AIM and PIM provide investors with a description of their behavior.